

Hilbert's

projective metric

Geometric Geometric

contraction coefficient that measures

spread/imbalance

Theorem [Sinkhorn Global Convergence]:

One has $(\mathbf{u}^{(\ell)}, \mathbf{v}^{(\ell)}) \rightarrow (\mathbf{u}^*, \mathbf{v}^*)$ and

$$d_{\mathcal{H}}(\mathbf{u}^{(\ell)}, \mathbf{u}^*) = O(\lambda(\mathbf{K})^{2\ell})$$

$$d_{\mathcal{H}}(\mathbf{v}^{(\ell)}, \mathbf{v}^*) = O(\lambda(\mathbf{K})^{2\ell})$$

$$\text{where } \lambda(K) = \frac{\sqrt{\eta(K)} - 1}{\sqrt{\eta(K)} + 1}, \quad \eta(K) = \max_{i,j,k,\ell} \frac{K_{i,k}K_{j,\ell}}{K_{j,k}K_{i,\ell}}.$$

On the Convergence Rate of Sinkhorn's Algorithm*

Promit Ghosal[†]

Marcel Nutz[‡]

April 8, 2025

Abstract

We study Sinkhorn's algorithm for solving the entropically regularized optimal transport problem. Its iterate π_t is shown to satisfy $H(\pi_t|\pi_*) + H(\pi_*|\pi_t) = O(t^{-1})$ where H denotes relative entropy and π_* the optimal coupling. This holds for a large class of cost functions and marginals, including quadratic cost with subgaussian marginals. We also obtain the rate $O(t^{-1})$ for the dual suboptimality and $O(t^{-2})$ for the marginal entropies. More precisely, we derive non-asymptotic



Sharper exponential convergence rates for Sinkhorn's algorithm in continuous settings

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Abstract

We study the convergence rate of Sinkhorn's algorithm for solving entropy-regularized optimal transport problems when at least one of the probability measures, μ , admits a

Sinkhorn: Convergence

Theorem [Sinkhorn Global Convergence]:

One has $(\mathbf{u}^{(\ell)}, \mathbf{v}^{(\ell)}) \rightarrow (\mathbf{u}^*, \mathbf{v}^*)$ and

$$d_{\mathcal{H}}(\mathbf{u}^{(\ell)}, \mathbf{u}^*) = O(\lambda(\mathbf{K})^{2\ell})$$

$$d_{\mathcal{H}}(\mathbf{v}^{(\ell)}, \mathbf{v}^*) = O(\lambda(\mathbf{K})^{2\ell})$$

$$\text{where } \lambda(K) = \frac{\sqrt{\eta(K)} - 1}{\sqrt{\eta(K)} + 1}, \quad \eta(K) = \max_{i,j,k,\ell} \frac{K_{i,k}K_{j,\ell}}{K_{j,k}K_{i,\ell}}.$$

- Convergence analysis greatly simplified using **Hilbert's projective metric** (a scale invariant metric $\rightarrow$ a natural choice here since $(\mathbf{u}, \mathbf{v})$ unique up to multipl. constant)
- **Guaranteed convergence:** Sinkhorn iterates $(\mathbf{u}^{(\ell)}, \mathbf{v}^{(\ell)})$ converge to the unique entropic OT solution.
- **Rate:** **Geometric** in Hilbert's projective metric.
- $\lambda(\mathbf{K}) < 1$ is a **contraction coefficient that measures spread/imbalance** of values in $\mathbf{K}$
- **Intuition:**
 - If rows of $\mathbf{K}$ are “balanced” (similar size) $\rightarrow \eta(\mathbf{K}) \gtrsim 1 \rightarrow \lambda(K)$ small $\rightarrow$ faster convergence
 - If rows of $\mathbf{K}$ are very uneven $\rightarrow \eta(\mathbf{K})$ large $\rightarrow \lambda(K) \lesssim 1 \rightarrow$ slower convergence
- Explains slow convergence of Sinkhorn for small ε : $K = e^{-C/\varepsilon}$ becomes peaked, uneven, $\lambda(K)$ worsens.

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FULL LENGTH PAPER

Series A

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Active area of research!

Hilbert Metric

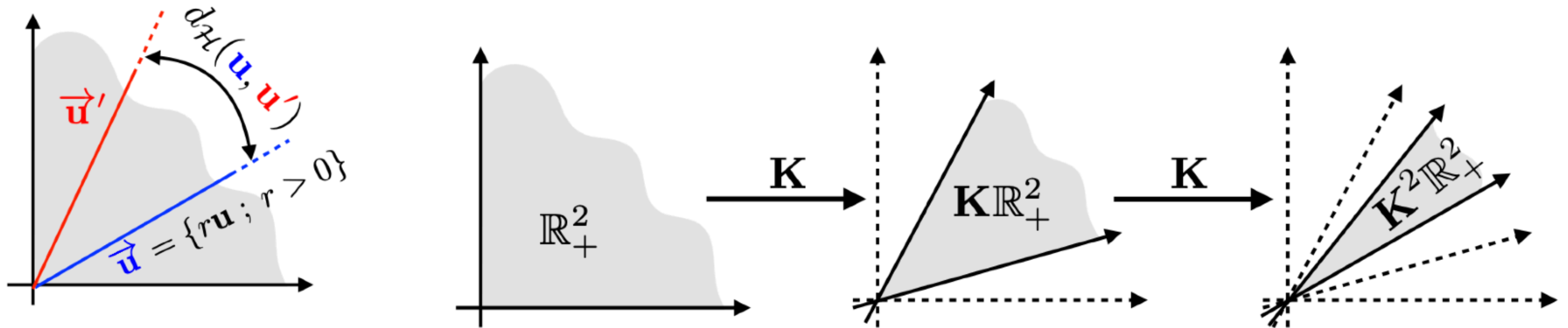


Figure 4.7: Left: the Hilbert metric $d_{\mathcal{H}}$ is a distance over rays in cones (here positive vectors). Right: visualization of the contraction induced by the iteration of a positive matrix $\mathbf{K}$.